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HOSPITAL MANAGEMENT SYSTEM (HMS) - SETUP & RUN STEPS
Stack: React.js + Laravel 11 + PHP 8.2 + MySQL
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PART 1: INSTALL REQUIRED SOFTWARE (one-time)
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STEP 1: Install XAMPP (gives you PHP + MySQL)

1. Download XAMPP with PHP 8.2 from: <https://www.apachefriends.org>
2. Install to C:\xampp
3. Add PHP to PATH:
 - Press Start, search "Edit the system environment variables"
 - Click "Environment Variables"
 - Under "System variables", select "Path" > Edit > New
 - Add: C:\xampp\php
 - Click OK on all windows
4. Enable PHP extensions:
 - Open C:\xampp\php\php.ini in Notepad
 - Find these lines and remove the leading semicolon (;) if present:
extension=pdo_mysql
extension=zip
extension=fileinfo
extension=openssl
extension=curl
 - Save the file

STEP 2: Install Composer

1. Download Composer-Setup.exe from: <https://getcomposer.org/download>
2. Run it - it will auto-detect PHP at C:\xampp\php\php.exe

STEP 3: Install Node.js

1. Download the LTS version from: <https://nodejs.org>
2. Install with default options (this also installs npm)

IMPORTANT: Close and reopen Command Prompt after installing,
so the new PATH takes effect. Verify with:

```
php -v
composer -V
node -v
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PART 2: START MYSQL & CREATE THE DATABASE
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STEP 4: Start MySQL

1. Open "XAMPP Control Panel"
2. Click "Start" next to MySQL (and Apache)

STEP 5: Create the database

1. Open <http://localhost/phpmyadmin> in your browser
2. Click "New" (left sidebar)
3. Database name: hms
4. Click "Create"

(XAMPP MySQL user is "root" with empty password -
this already matches the project's .env defaults)

PART 3: RUN THE BACKEND (Laravel API)

STEP 6: Open Command Prompt and run:

```
cd C:\Users\SURAJIT CHAKRABORTY\Downloads\hms\backend
composer install
copy .env.example .env
php artisan key:generate
php artisan migrate --seed
php artisan serve
```

Backend now runs at: <http://localhost:8000>
KEEP THIS WINDOW OPEN.

PART 4: RUN THE FRONTEND (React)

STEP 7: Open a SECOND Command Prompt and run:

```
cd C:\Users\SURAJIT CHAKRABORTY\Downloads\hms\frontend
npm install
npm run dev
```

Frontend now runs at: <http://localhost:5173>
KEEP THIS WINDOW OPEN TOO.

PART 5: LOG IN

Open <http://localhost:5173> in your browser.

Demo accounts (password for all: password)

Role	Email
Admin	admin@hms.test
Doctor	anil@hms.test
Receptionist	reception@hms.test
Patient	patient@hms.test

New patients can also self-register from the login page.

DAILY USE (after first-time setup)

1. Start MySQL in XAMPP Control Panel
2. Terminal 1: `cd ...\hms\backend -> php artisan serve`
3. Terminal 2: `cd ...\hms\frontend -> npm run dev`
4. Open <http://localhost:5173>

TROUBLESHOOTING

- "composer/php/node is not recognized"
 - > Reopen Command Prompt; check PATH (Part 1, Step 1.3)
- "could not find driver" during migrate
 - > Enable extension=pdo_mysql in php.ini (Part 1, Step 1.4)
- "Access denied for user 'root'"
 - > Check DB_USERNAME=root and DB_PASSWORD= (empty) in backend\.env
- "Unknown database 'hms'"
 - > Create the database in phpMyAdmin (Part 2, Step 5)
- Port 3306 already in use
 - > Another MySQL is running; stop it or change the port
- Frontend shows network errors
 - > Make sure "php artisan serve" is still running on port 8000